Summary in Graph

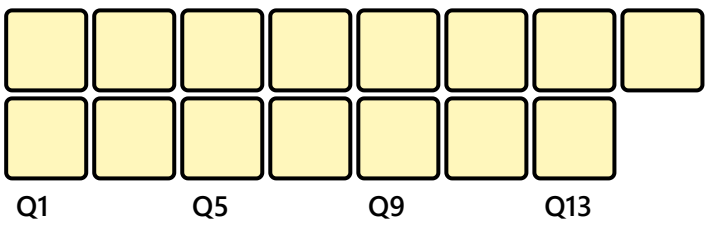
Exam Summary (GO Classes GATE CS | Operating System | Multi Level Paging | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (19) out of 192 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Operating System

Suppose that pages are 512B and each page table entry is 4 bytes.
Assume that somehow the virtual and physical address spaces were both 4GB and that the page table begins at address $0\text{x}10000000$. If we wanted to access the virtual address $0\text{x}00000345$, what is the address of the PTE we would look at?

A. 0x10000000

B. 0x10000001

C. 0x10000004

D. 0x10000345

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Select all True statements.

A. A process's virtual address space can be larger than the size of the physical address space.

B. The physical address space can be larger than a process's virtual address space.

C. A process cannot access more memory than the size of its virtual address space.

D. A process cannot access more memory than the size of the physical address space.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A processor uses 11-bit physical addresses and 16-bit virtual addresses.

The page size is 2^8 bytes, and a two-level paging scheme is used for address translation.

The virtual address is divided into the following fields:

5 bits	3 bits	8 bits
L1 index	L2 index	Page offset

Each Page Table Entry is 2 bytes in size and has the following format:

Physical Page Number (PPN)	Control bits

Determine which of the following L1 and L2 PPN values would NOT be possible:

A. L1 PPN value $= 120$, L2 PPN value $= 5$

B. L1 PPN value $= 64$, L2 PPN value $= 3$

C. L1 PPN value $= 127$, L2 PPN value $= 7$

D. L1 PPN value $= 12$, L2 PPN value $= 15$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Numerical Type | Award: 1 | Penalty: 0 | Operating System

A system uses a 47-bit virtual address space with a page size of $\mathbf{16\ KB}$. Each page table entry (PTE) requires 8 bytes. Each page table must fit entirely within a single page. How many levels of page tables are required to map the entire virtual address space?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A system uses a linear page table for a virtual address space of size 128 bytes with 32-byte pages.
Each Page Table Entry (PTE) contains the Page Frame Number (PFN) followed by a one valid bit.
The page table entries are as follows:

VPN	PTE (Hex)
0	0xFE
1	0x0F
2	0xFE
3	0x09

Now, for the following virtual addresses, determine which ones are valid (no page fault):

A. $0\text{x}065$

B. $0\text{x}00\text{C}$

C. $0\text{x}026$

D. $0\text{x}058$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A system uses 15-bit virtual addresses, a 32-byte page size, and a two-level paging scheme.
The address bits (other than the offset) are split equally between the outer and inner page table indices.
Each Page Table Entry (PTE) consists of a valid bit (MSB) followed by a 7-bit Page Frame Number (PFN).
The outer page table is located in physical page 73.

Partial physical memory (in hexadecimal, 0-based indexing):

```
pg 6 :  0a 1c 01 14 0b 1a ...
pg 73 :  a2 d2 97 96 f2 7f ...
pg 114 :  7f 7f 7f 86 1f 0c ...
```

The virtual address to be translated is $0\text{x}1787$.
What is the value read from memory after translation?

A. $0\text{x}7F$
B. $0\text{x}86$
C. $0\text{x}1A$
D. Page fault occurs

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

A computer system implements a two-level paging scheme with the following parameters:

- Virtual address size: 42 bits
- Physical address size: 40 bits
- Page size: $64\ \mathrm{KB} = 2^{16}$ bytes
- Each page table (at both levels) occupies exactly one page (64 KB).
- Each Page Table Entry (PTE) occupies 8 bytes

The system accesses the virtual address $0\text{x}\ 00\ 12\ 34\ 56\ 78\ 9$.
Select all virtual addresses from the following that use the same outermost page table entry as this address:

A. $0\text{x}\ 00\ 12\ 34\ 44\ 44\ 4$
B. $0\text{x}\ 00\ 0A\ 98\ 76\ 78\ 9$
C. $0\text{x}\ 00\ 13\ 63\ 45\ 67\ 8$
D. $0\text{x}\ 00\ 00\ 00\ 06\ 78\ 9$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 01sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A system uses 32-bit virtual addresses and 4-KB pages.
A three-level page table is used as follows:

- A single root page directory points to multiple mid-level directories.
- Each mid-level directory points to page table pages that store the actual page table entries (PTEs).

Assume that only two pages in the virtual address space are valid: the very first page and the very last page.
How many total physical pages are used to store this page table structure (including all levels of indirection)?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A process has a 32-bit virtual address space and uses two-level paging with a 4 KB page size. Assume the process accesses one page at the top, one in the middle, and one at the bottom of its virtual address space. How large (in bytes) will the page table structure be (counting only the page tables, not data pages)?

A. 2^{12} bytes

B. 2^{13} bytes

C. 2^{14} bytes

D. 2^{15} bytes

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A system uses 34-bit virtual addresses and two-level paging.
The page size is 2 KB (i.e., offset $= 11$ bits).
The virtual address is divided as:

n bits	$(34 - n - 11)$ bits	11 bits
L1 index	L2 index	Page offset

A program uses 8 MB of virtual memory.
What is the minimum number of second-level page tables required to map the entire address space used by the program?

A. 2^{n-12}

B. $2^{(n-11)}$

C. $2^{(34-n-12)}$

D. $2^{(n-23)}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Suppose you search for an element x in a sorted array of N elements using a linear search.
Each memory page holds p array elements ($N > p$).
Assume that the cost of a page fault is much higher than that of a normal memory access.
On average, how many page faults will be generated during the linear search?

A. $\frac{N}{2}$

B. $\frac{N}{2p}$

C. $\log \frac{N}{p}$

D. $\frac{p}{N}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Suppose you search for an element x in a sorted array of N elements using a binary search.
Each page holds p array elements ($N > p$).
Assume that the cost of a page fault is much higher than that of a normal memory access.
On average, how many page faults will be generated during the binary search?

A. $\frac{\log N}{p}$

B. $\log \frac{N}{p}$

C. $\frac{N}{2p}$

D. $\frac{p}{N}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A processor uses two-level page tables for virtual-to-physical address translation. Both levels of page tables are stored in the main memory. Virtual and physical addresses are 32 bits wide, and the system uses 4 KB pages with 4-byte page table entries (PTEs).
The two-level split is such that each page table contains the same number of entries. Memory is byte-addressable.
A program uses only the following virtual addresses during its execution:

0x00001000

0x00002000

0xFFFFFE00

0xFFFFFF00

Assume that a page table is allocated only when needed and that each page table occupies one full page frame.
The total memory required to store all page table structures for this program is:

A. 4 KB

B. 8 KB

C. 12 KB

D. 16 KB

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a virtual memory system that uses 2-level paging. The page size in this system is 256 bytes. Each individual page table fits exactly into one memory frame, and the size of each page table entry (PTE) is 8 bytes.
Suppose that there is a process with a virtual address space of size 10240 $\left(10\text{x}2^{10}\right)$ bytes. Also suppose that the entire address space is in memory.
How many valid PTEs will exist in the page tables for this process?

A. 32

B. 40

C. 42

D. 48

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A system uses 15-bit virtual addresses, a 32-byte page size, and a two-level paging scheme.
Assume both levels have equal bits in address split.
Each outer and inner page table entry is **1** byte (MSB = valid bit, remaining **7** bits store the frame number).
Physical memory has **128** pages.

The relevant portion of physical page **93** is shown below:

```
pg 93 :  15 08 08 00 11 15 07 02 12 12 0c 03 1c 1a 13 11 ...
```

If the virtual address that refers to a byte on this page is **0x48CB**, which byte value would be returned by a load to that address?

A. 0x0C
B. 0x02
C. 0x03
D. 0x1A

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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